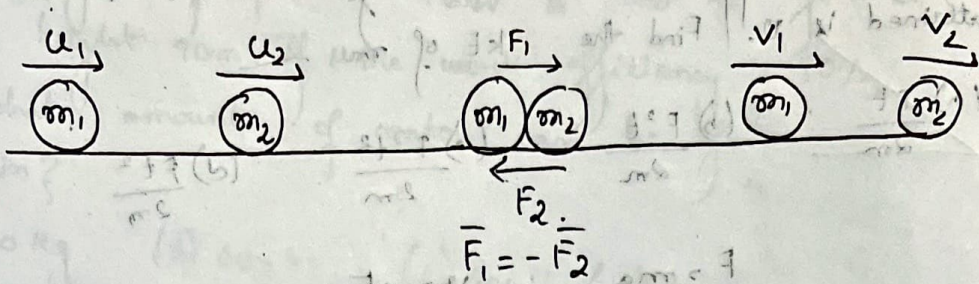


* Collision - (Elastic)



$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2 \quad (\text{law of conservation of momentum})$$

$$\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 \quad (\text{law of conservation of energy})$$

from the 1st eqⁿ

$$m_1 u_1 - m_1 v_1 = m_2 v_2 - m_2 u_2$$

$$\Rightarrow m_1 (u_1 - v_1) = m_2 (v_2 - u_2) \quad \text{--- (1)}$$

from the second eqⁿ

$$\frac{1}{2} m_1 (u_1^2 - v_1^2) = \frac{1}{2} m_2 (v_2^2 - u_2^2)$$

$$\Rightarrow m_1 (u_1 + v_1) (u_1 - v_1) = m_2 (v_2 - u_2) (v_2 + u_2) \quad \text{--- (2)}$$

$\therefore$ dividing (2) by (1)

$$\therefore \frac{m_1 (u_1 + v_1) (u_1 - v_1)}{m_1 (u_1 - v_1)} = \frac{m_2 (v_2 - u_2) (v_2 + u_2)}{m_2 (v_2 - u_2)}$$

$$\therefore \boxed{u_1 + v_1 = v_2 + u_2}$$

$$v_1 = v_2 + u_2 - u_1$$

Putting the above value in the 1st eqⁿ

$$m_1 u_1 + m_2 u_2 = m_1 (v_2 + u_2 - u_1) + m_2 v_2$$

$$\Rightarrow m_1 u_1 + m_2 u_2 = m_1 v_2 + m_1 u_2 - m_1 u_1 + m_2 v_2$$

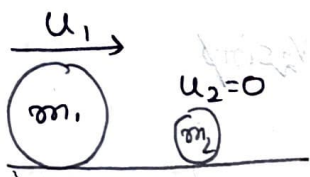
$$\Rightarrow 2m_1 u_1 + (m_2 - m_1) u_2 = (m_1 + m_2) v_2$$

$$\therefore v_2 = \frac{2m_1}{m_1 + m_2} u_1 + \frac{m_2 - m_1}{m_1 + m_2} u_2$$

$$\therefore v_1 = \frac{2m_2}{m_1 + m_2} u_2 + \frac{m_1 - m_2}{m_1 + m_2} u_1$$

In elastic collision there is no loss of energy.

* Energy loss due to partial elastic collision-



Before collision the total energy of the system.

$$E_1 = \frac{1}{2} m_1 u_1^2 + 0 = \frac{1}{2} m_1 u_1^2$$

After collision the total energy of the system.

$$E_2 = \frac{1}{2} (m_1 + m_2) v^2$$

$$v = \frac{m_1 u_1}{m_1 + m_2}$$

$$\therefore E_2 = \frac{1}{2} (m_1 + m_2) \left(\frac{m_1 u_1}{m_1 + m_2} \right)^2$$

$$= \frac{1}{2} \frac{m_1^2 u_1^2}{m_1 + m_2}$$

$$\therefore \text{Loss of energy} = E_1 - E_2$$

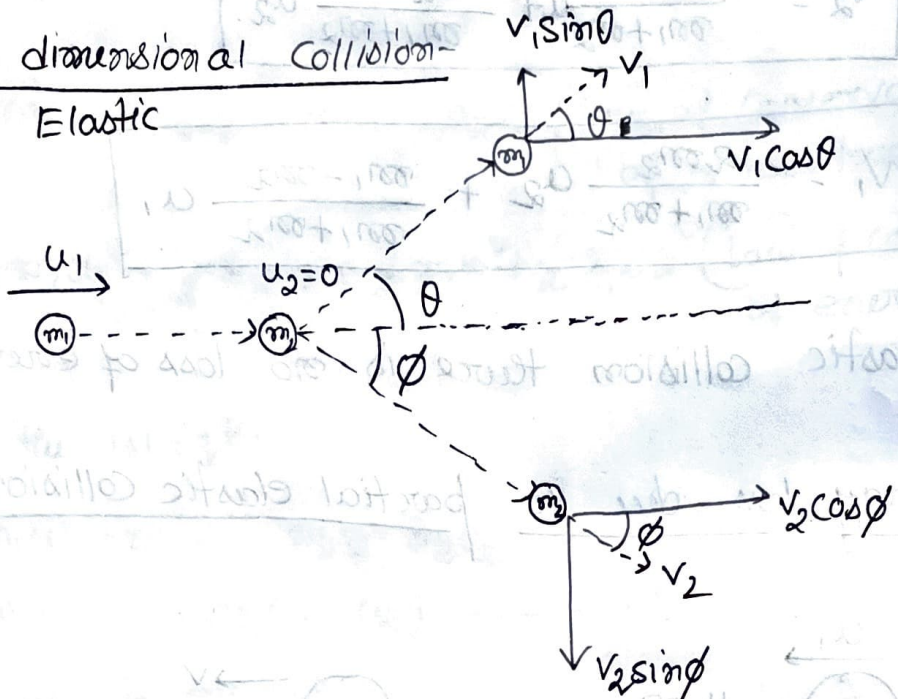
$$= \frac{1}{2} m_1 u_1^2 - \frac{1}{2} \frac{m_1^2 u_1^2}{m_1 + m_2}$$

$$= \frac{1}{2} u_1^2 \left[m_1 - \frac{m_1^2}{m_1 + m_2} \right]$$

$$\Delta E = \frac{1}{2} u_1^2 \left[\frac{m_1 + m_2 - m_1}{m_1 + m_2} \right]$$

$$\Delta E = \frac{1}{2} \left(\frac{m_1 m_2}{m_1 + m_2} \right) u_1^2$$

* Two dimensional collision
Elastic



∴ the law of conservation of momentum in horizontal direction-

$$m_1 u_1 + 0 = m_1 v_1 \cos \theta + m_2 v_2 \cos \phi$$

$$\therefore m_1 u_1 = m_1 v_1 \cos \theta + m_2 v_2 \cos \phi$$

∴ the same for the vertical direction-

$$0 = m_1 v_1 \sin \theta - m_2 v_2 \sin \phi$$

$$m_1 v_1 \sin \theta = m_2 v_2 \sin \phi$$

As the collision is elastic so it also obeys the law of conservation of energy.

$$\frac{1}{2} m_1 u_1^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$

$$\therefore m_1 u_1^2 = m_1 v_1^2 + m_2 v_2^2$$